Balance each direction, not just the force sizes

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Equilibrium / ES: equilibrio / PT: equilíbrio

Zero a : forces sum to zero on each axis.

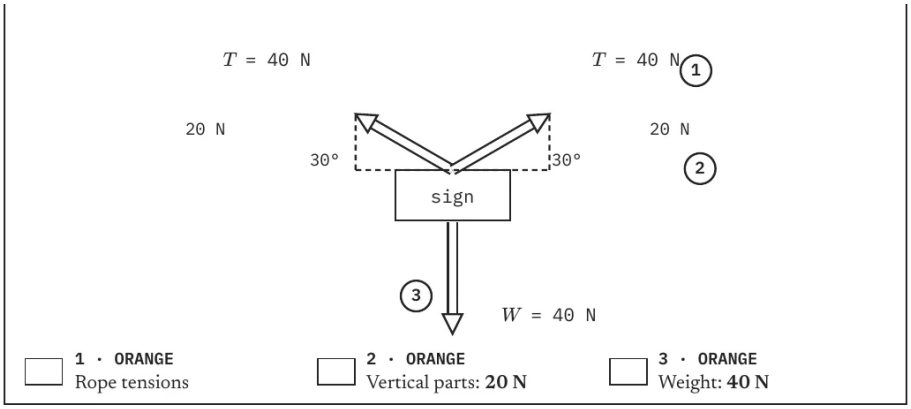
Also on your note page: Balance, Symmetry, Rest.

B READ THE RELATIONSHIP

Equilibrium requires zero force sum on every axis

As you read, find: balance · upward part · weight.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 40 N sign hangs at rest from two equal ropes, each 30° above horizontal. Find the tension in each rope.

Given W = 40 N ; θ = 30° equal tensions a = 0

Rule x parts cancel. 2 T sin 30° = 40 N .

Work 2 T (0.5) = 40 N T = 40 N

Answer 40 N per rope Each rope gives 20 N upward.

E SAVE FOR THE FINAL MEETING

Zero net force would give XJ a constant velocity, not a guaranteed stop. Analysts must check each coordinate direction before treating its motion as steady.

Read the physics: Equilibrium

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Choose horizontal and vertical axes. Add the forces on each axis. When both sums are zero, acceleration is zero. The object can be at rest or moving steadily.

02 HOW IT WORKS

A hanging sign has downward weight and upward support. If the ropes are vertical, add their tensions. If the ropes are tilted, only the upward parts support the weight.

03 WHAT TO CHECK

Being at rest for an instant is not enough to prove equilibrium. Look for zero net force and unchanged velocity.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 40 N sign hangs at rest from two equal ropes, each 30° above horizontal. Find the tension in each rope. 40 N per rope Each rope gives 20 N upward.

Model check: Equilibrium requires zero force sum on every axis.

05 CONNECT TO XJ

Zero net force would give XJ a constant velocity, not a guaranteed stop. Analysts must check each coordinate direction before treating its motion as steady.

SAY IT BEFORE YOU WRITE

Can an object in equilibrium be moving? The ___ forces balance. Its acceleration is ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Equilibrium / ES: equilibrio / PT: equilíbrio

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, equilibrium means …	A cart rolls straight at steady speed. What must the forces on it add up to? The forces add up to ___ because the velocity ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch two systems with balanced forces: one at rest and one moving steadily. Label the common acceleration.

Labels you can use: balance · upward part · weight.

At rest

Moving steadily

Tell a partner: The ___ forces balance. Its acceleration is ___.

On assignment: Equilibrium

SECTION E / BACK / FINAL MEETING ONLY Name _____

The flight display reads “net force zero.” XJ’s last velocity was +3 km/s. Two graphics say STOPPED and COASTING.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Flight controller**

Select the status graphic. Plot a second position two seconds later relative to the first and explain your instruction.

☐ **Broadcast producer**

Choose the lower-third for a live shot. Ask the anchor to distinguish zero acceleration from zero speed before airing it.

Word bank: balance · upward part · weight.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: Check the claimed balance along both coordinate axes.